

$$\begin{aligned}
 \int_0^\infty \varphi(y) dy &= \int_0^\infty dy \int_E f(x) \chi_{E(g \geq y)}(x) dx \\
 &= \int_E dx \int_0^\infty f(x) \chi_{E(g \geq y)}(x) dy \\
 &= \int_E dx \int_0^{g(x)} f(x) dy = \int_E f(x) g(x) dx.
 \end{aligned}$$

13. 因为 $\int_{E \times E} (f(x) + g(y)) dx dy = \int_E dx \int_E (f(x) + g(y)) dy$, 所以对于几乎处处 $x \in E$, 有 $f(x) + g(y) \in L(E)$. 又 $m(E) < \infty$, 因而 $g(y) \in L(E)$. 同理 $f(x) \in L(E)$.

14. (1) $\pi^2/2$; (2) 0.

15. 用反证法.

16. 因为 $c \leq f(x) \leq d$, $y_0 = \int_E f(x) g(x) dx \in [c, d]$, 所以有

(1) 设 $y_0 \in (c, d)$. 由函数 φ 的凸性, $\exists k$, 使得

$$\varphi(y) \geq \varphi(y_0) + k(y - y_0); \quad \varphi(f(x)) \geq \varphi(y_0) + k(f(x) - y_0);$$

$$\int_E \varphi(f(x)) g(x) dx \geq \varphi(y_0) + k \left(\int_E f(x) g(x) dx - y_0 \right) = \varphi(y_0).$$

(2) 设 $y_0 = d$ (或 c), 则有

$$\int_E (d - f(x)) g(x) dx = 0,$$

$\Rightarrow$

$$f(x) = d, \quad \text{a.e.},$$

$\Rightarrow$

$$\int_E \varphi(f(x)) g(x) dx = \varphi(d) \int_E g(x) dx = \varphi \left(\int_E f(x) g(x) dx \right).$$

17. $\sqrt{fg} \geq 1$, 考虑

$$0 \leq I(\lambda) \stackrel{\text{def}}{=} \int_E (\sqrt{f(x)} + \lambda \sqrt{g(x)})^2 dx = a\lambda^2 + 2b\lambda + c,$$

其中

$$a = \int_E g dx, \quad b = \int_E \sqrt{fg} dx, \quad c = \int_E f dx.$$

判别式小于等于零,

$$1 \leq \left(\int_E \sqrt{f(x)g(x)} dx \right)^2 \leq \int_E f(x) dx \int_E g(x) dx.$$